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SPECTRALLY ENCODED TWO-VARIABLE MAPS*

JUDY M. OLSON

ABSTRACT. The U.S. Census Bureau began publishing spectrally encoded two-variable choropleth maps in the early 1970s. Four experiments with human subjects have been conducted with these controversial maps to examine their efficacy. It has been shown that 1) subjects cannot spontaneously order the colors into the legend arrangement used on these maps, but they can recognize order in that arrangement; 2) the task of judging correlation between the geographical patterns of individual variables is initially more difficult than the task of judging correlation on separate maps, although subjects who do better than guessing on this task perform at least as well as they do with separate maps; 3) subjects do show evidence of having gained information when they are asked to write what they learn from these maps; and 4) subjects find the maps interesting and appealing, particularly if explanatory verbal notes accompany them.

THE choice of mapping method has a profound effect upon the information that is available from a map. Although the choice of method is largely within the province of the map*maker*, it is the effect of the method on the *user* that ultimately determines the success or failure of the choice. Further, the user makes implicit choices about mapping methods when deciding whether and how to utilize available maps. The efficacy of mapping methods, then, is a question of general interest to cartographers, geographers, and others whether they make maps or use them.

The mapping method under scrutiny in this study is the spectrally encoded two-variable choropleth technique (Fig. 1). This technique is a variation of simple choropleth mapping, in which the value for each data collection unit is represented by a tone or pattern covering the entire unit.[1] In the two-variable case the tones representing values on two different distributions are overprinted and the resulting tones represent combinations of values on the two distributions (Figs. 2 and 3). When the tones representing the two distributions are composed of all three primary colors, the maps can be referred to as "spectrally encoded."

Two-variable choropleth mapping undoubtedly has a history that goes back far beyond the present, but it gained special prominence in the early 1970s when the U.S. Bureau of the Census employed its automated mapping system to produce several such maps for widespread distribution (Figs. 1, 4, and 5).[2] The spectrally en-

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[1] The precise definition of choropleth is not universally agreed upon in cartography. Any map representing a value averaged over a data collection unit (say, population density by county) and represented by a symbol covering the entire unit seems consistently to be included within the definition; inconsistency of definition arises when considering the representation of absolute values (e.g., total population in each county), uniform values over areal units (e.g., tax rates), or any map representing a value averaged over the unit but employing an alternative symbol system (e.g., graduated symbols).

[2] For descriptions of the production system, see Morton A. Meyer, Frederick R. Broome, and Richard H. Schweitzer, Jr., "Color Statistical Mapping by the U.S. Bureau of the Census," *The American Cartographer*, Vol. 2 (1975), pp. 100–17; and Frederick R. Broome, "Micrographics: A New Approach to Cartography at the Census Bureau," *Proceedings of the Fall Convention*, American Congress on Surveying and Mapping, September 1974, pp. 1–14. The maps produced included several of the U.S. by county. In addition to those in Figs. 1, 4, and 5 were "Convergence of Male Cardiovascular Disease Mortality and Percent With 1.01 or More Persons Per Room, 1968–1971" and "Combination of Percent of Population 65 Years Old and Over 1975, and Per Capita Income, 1974," which appeared in the August and September issues, respectively, of the short-lived publication: U.S. Bureau of the Census, *STATUS: A Monthly Chartbook of Social and Economic Trends* (Washington, D.C.: U.S. Government Printing Office, 1976). A two-variable map of education and income by census tract appeared in each volume of the Urban Atlas series: U.S. Bureau of the Census and Manpower Administration, *Urban Atlas*, GE80 series (Washington, D.C.: U.S. Government Printing Office, 1974).

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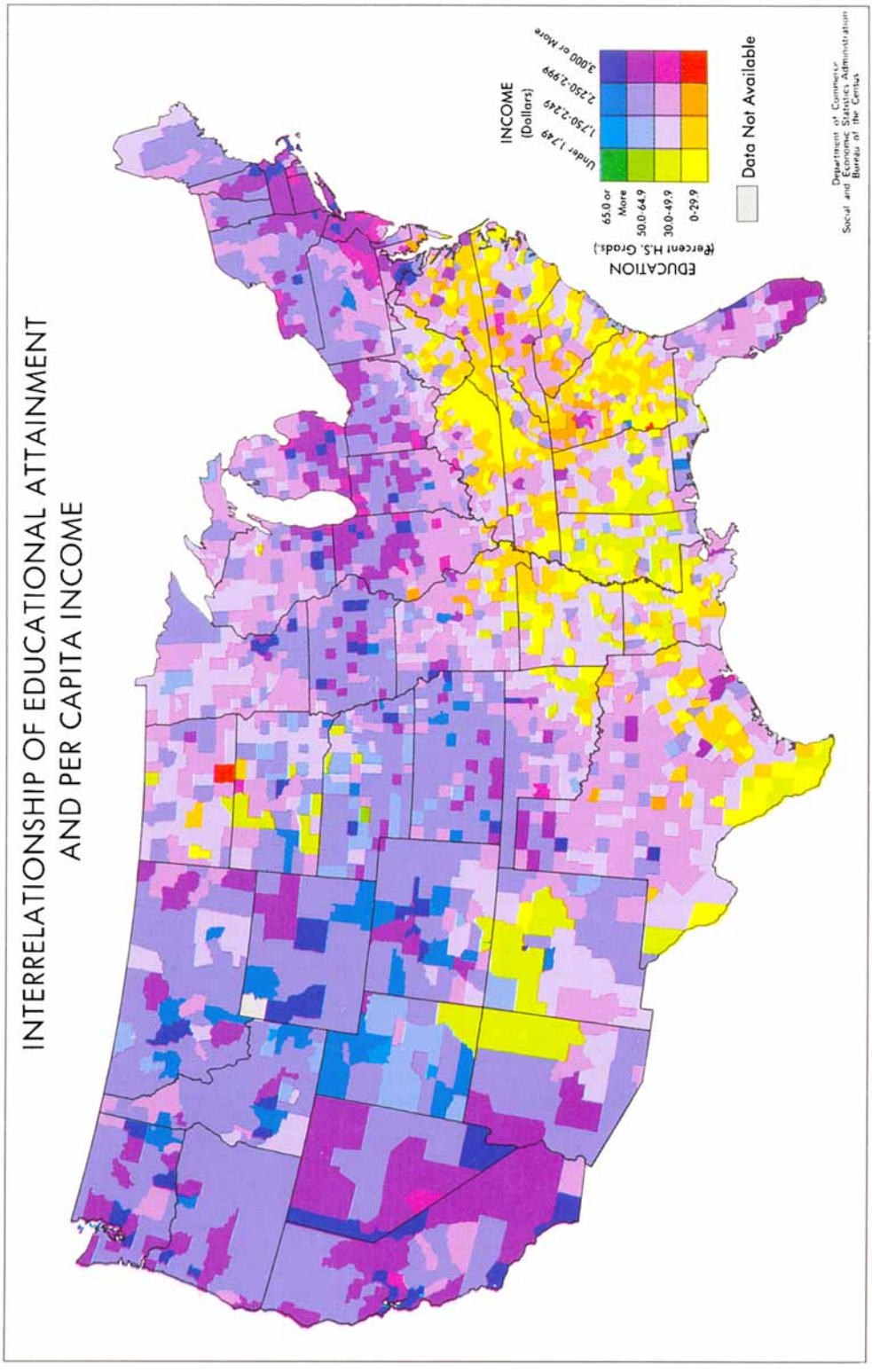

INTERRELATIONSHIP OF EDUCATIONAL ATTAINMENT
AND PER CAPITA INCOME
EDUCATION
(Percent H.S. Grads.)
65.0 or More
50.0-64.9
30.0-49.9
0-29.9
INCOME
(Dollars)
Under 1,749
1,750-2,249
2,250-2,999
3,000 or More
Data Not Available
Department of Commerce
Social and Economic Statistics Administration
Bureau of the Census

FIGURE 1

coded system of symbolization was used on these Census maps.[3]

The Bureau's two-variable products have generated numerous adverse criticisms from cartographers and others with interests in maps, primarily at professional meetings and in informal professional conversation. While it would be difficult at best to report in detail on controversy which is largely unpublished, one general observation is that these criticisms are based on very specific notions about what such a map should show. The lack of visual separability of the two distributions is commonly mentioned, and the statement that "not enough effort has gone into making them show bivariate relationships" reflects another line of argument.[4]

Notions about what should be visible on these maps reflects, in turn, a general line of thinking about mapping that says that the map maker has, or should have, a specific message in mind when making a map and that message should be conveyed to the map user. There is considerable credibility in such a view; surely one would not suggest that what the map user receives has no relationship to what was in the mind of the cartographer. Yet the popular viewpoint also constrains thinking about the evaluation of maps in that it suggests a research question such as the following: In constructing the two-variable map the mapmaker has taken two undulating surfaces and superimposed them; does the map user see two intersecting, undulating surfaces?

Fortunately, there is also considerable recognition in the field that the map user may be reading, analyzing, or interpreting things of which the cartographer was not aware.[5] This recognition gives special significance to the idea of "the map user's point of view" and it suggests that users need not be evaluated on the basis of how well they duplicate the line of thinking taken by the mapmaker in producing the product. Questions can then be asked about whether the map user can gain any sort of logical information from a map, some of which may be very much related to the mapmaker's thinking, some of which may not.

The questions used in this study reflect, in part, an effort to approach map evaluation from the viewpoint of map use, rather than from the notion that user thinking should duplicate that of the mapmaker. The testing also reflects that it was the *mapping method* being evaluated, not a specific symbol or the map of a specific phenomenon. Further, it reflects the assumption that there are several essential components to the use of maps; a mapping method should not be evaluated on the basis of some one isolated criterion. With these thoughts in mind, four different experiments were chosen for inclusion, and they concerned: 1) the intelligibility of the symbol system (to the map user), 2) intelligibility of the spatial pattern, or geographical arrangement, of the symbols (to the user), 3) the ability of users to extract information without being questioned about specific parts or relationships, and 4) map user attitudes toward the maps.

Experiment 1—The Symbol System

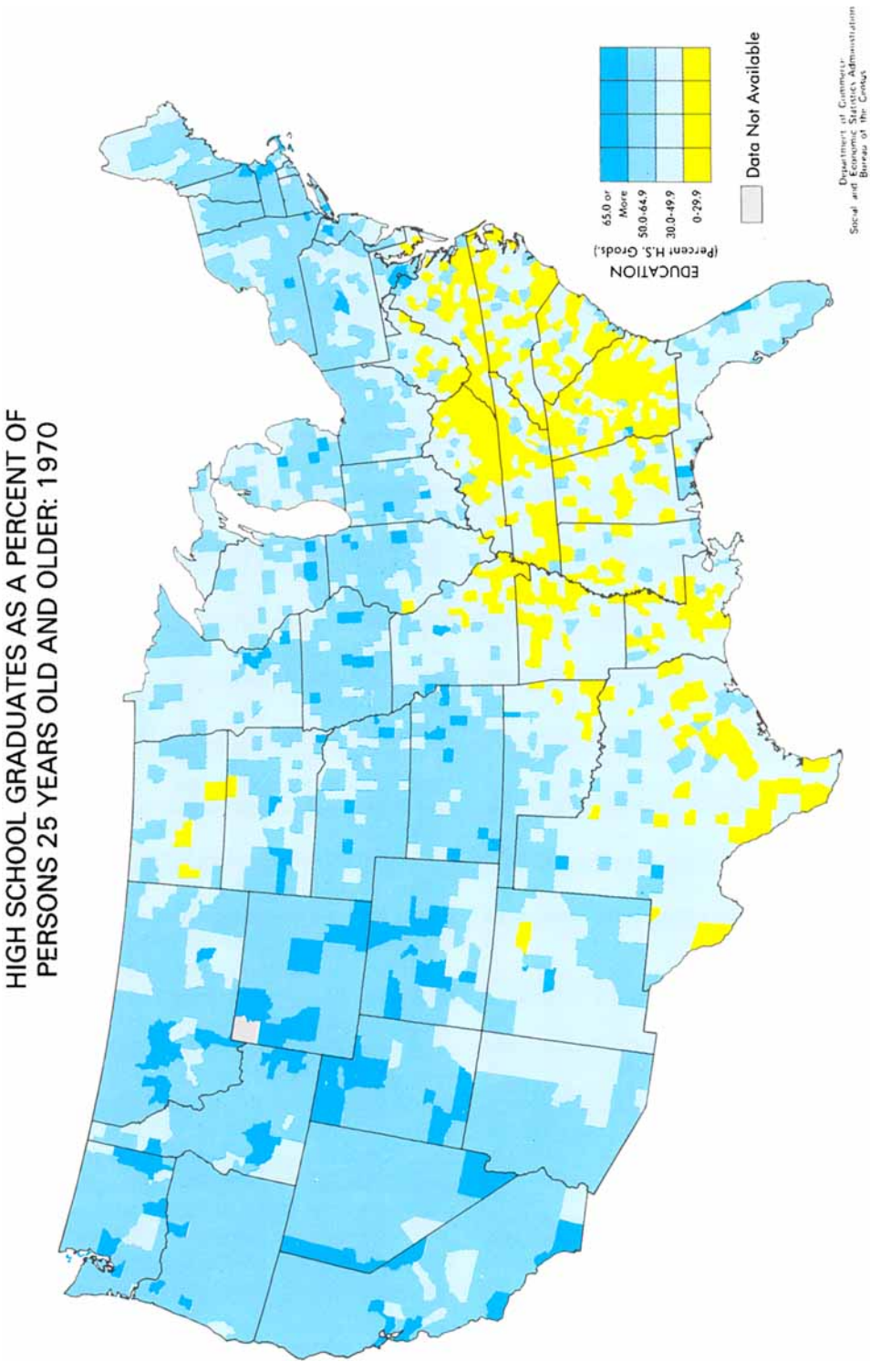
One of the most important relationships encoded within a quantitative map is that between the symbol system and the relative values represented. "Relative values" in this context refers to their order with respect to one another—more than, less than—as opposed to actual numerical values represented. With some map symbols the relative relationship is so straightforward and universal that it is taken for granted; for example, it is simply assumed that a greater density of dots on a dot map indicates greater density of the phenomenon represented and that a larger symbol on a graduated symbol map represents a larger value. The symbol-value relationship is not so straightforward when hue is a variable in the symbol system because hue

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[3] This method is explained in Vincent P. Barabba, "The Graphic Presentation of Statistics," pp. 89–103 in U.S. Bureau of the Census, *The Census Bureau: A Numerator and Denominator for Measuring Change*, Technical Paper 37 (Washington, D.C.: U.S. Government Printing Office, 1975).

[4] Stephen E. Feinberg, "Graphical Methods in Statistics," *The American Statistician*, Vol. 33 (1979), p. 176. The many considerations one might keep in mind in attempting to design such a color system are outlined in Judy M. Olson, "The Organization of Color on Two-Variable Maps," *Proceedings of AUTO-CARTO II* (Falls Church, Va.: the American Congress on Surveying and the U.S. Bureau of the Census, 1975), pp. 289–94 (with figures on pp. 251, 264–66). A recently published paper that included testing of human subjects is Howard Wainer and Carl M. Francolini, "An Empirical Inquiry Concerning Human Understanding of Two-Variable Color Maps," *The American Statistician*, Vol. 34 (1980), pp. 81–93.

[5] Joel L. Morrison, "A Theoretical Framework for Cartographic Generalization with Emphasis on the Process of Symbolization," *The International Yearbook of Cartography*, Vol. 14 (1974), p. 117.


HIGH SCHOOL GRADUATES AS A PERCENT OF PERSONS 25 YEARS OLD AND OLDER: 1970
EDUCATION
(Percent H.S. Grads.)
65.0 or More
50.0-64.9
30.0-49.9
0-29.9
Data Not Available
Department of Commerce
Social and Economic Statistics Administration
Bureau of the Census

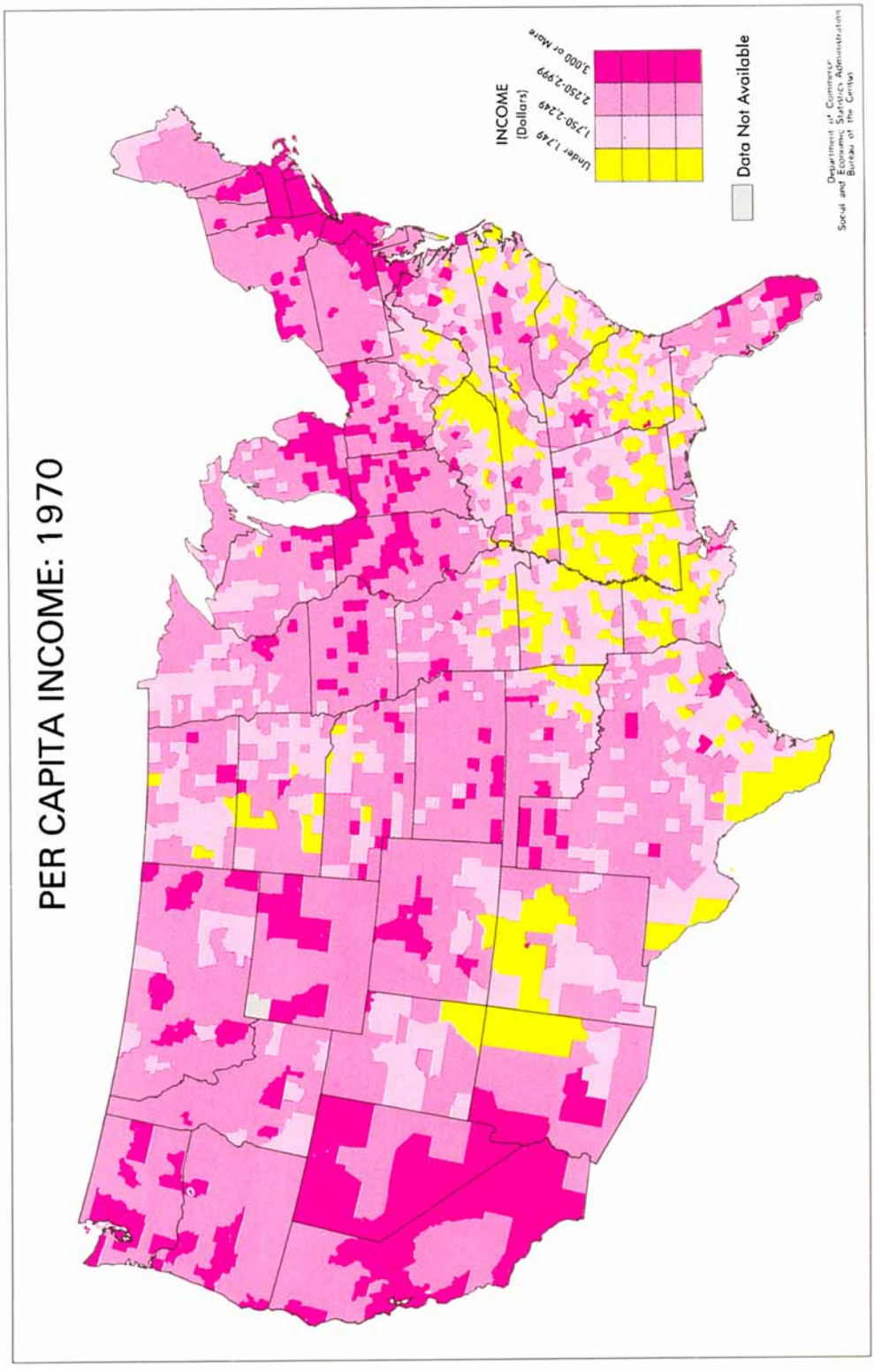

PER CAPITA INCOME: 1970
INCOME
(Dollars)
Under 1,749
1,750-2,249
2,250-2,999
3,000 or More
Data Not Available
Department of Commerce
Social and Economic Statistics Administration
Bureau of the Census

FIGURE 3

variation is generally associated with qualitative rather than quantitative distinctions. The question, then, is whether the assignment of such tones to value combinations is consonant with spontaneous visual association among the colors.

The testing included both three- and four-class schemes, since the answer to the question could well be affected by the number of tones used (Fig. 6). Three and four classes were chosen because it was reasoned that the mapping of quantitative variables should involve at least three categories while five (resulting in twenty-five combination tones) would not be feasible on these maps. The color sets used in Experiment I differ somewhat from those used by the Census Bureau in that yellow has been eliminated from the tones representing one of the variables and yellows and reds have been blended for the middle class(es) of the other variable. The modifications eliminate the block of very similar colors in the upper right of the legend arrangement and there is a somewhat more even transition among neighboring colors. It was assumed, however, that the spectral nature of the tones was a far more important influence on visual associations than any minor variations in the colors composing the spectral set.

The test sheet for Experiment I had an empty legend box and subjects were instructed to place either nine or sixteen color chips into the legend matrix such that they "best . . . matched . . . the value combinations" (Fig. 7).[6] The color chips were dumped from an envelope onto the test sheet, so they were in no particular order. The grids into which subjects were to place the chips were labeled high, medium, and low, and one axis was marked education, the other income.[7]

Subjects were tested individually or in small groups. Fourteen subjects worked with the nine-color set and thirteen with the sixteen-color set; all subjects were students in an Introductory Psychology class at Boston University and most took about five minutes to complete the task.

Results of the test might have been 1) no or few subjects arranging the tones differently from the original scheme, indicating that on the average this arrangement coincides with our mental ordering of these colors; 2) some other arrangement being chosen by all or nearly all subjects, in which case this alternative would be a more appropriate system to use; or 3) total or near total disagreement between subjects' arrangements, in which case there would be no evidence of a spontaneously associable arrangement. There was no doubt that the results fell into the third category of possibilities; no two subjects produced the same arrangement. There were, to be sure, some threads of similarity, such as yellow always appearing as an outer color rather than being placed in the middle boxes, but there was no arrangement of all nine or sixteen colors that could be considered as optimal on the basis of spontaneous association.

Following the main task, several questions were asked of subjects to avoid naive interpretation of results. These were conversational rather than precisely preworded but they were always given in the same order and were approximately as follows:

1) To the extent you can verbalize about it, tell why you arranged the colors as you did.

2) Does your arrangement look logical now or are you dissatisfied with it? (Did you find yourself wanting a different set of colors to work with?)

3) Were some colors more difficult to place than others?

4) This is another possible arrangement (showing them the original arrangement, Fig. 6); does this look like a logical arrangement? How would you describe it?

5) If you were actually working with or making maps of this sort, which arrangement would you use, the one I have shown you or your own?

By asking the various follow-up questions, it was soon revealed that subjects were *not*, in

[6] Exact wording of instructions was as follows: "You have been provided a set of nine (or sixteen) color patches. Assume these colors are to be used on a map to represent value combinations of two variables such as education and income. How do you think they should best be matched with the value combinations? Answer by placing the patches in the squares *below*. Let me know as soon as you are satisfied with the arrangement. The maximum time allowed will be 10 minutes. Turn the color side of the patches up before you begin placement."

[7] Avoiding the variable names on the test sheet would have been desirable so that associations could be strictly between symbol system and relative values represented, but for practical reasons the labeling was necessary. Value labels alone (high, medium, low) would not have made the task meaningful to subjects with no particular awareness of quantitative graphic symbolization. Thus to avoid long, tedious, and vague instructions, the variable names were included.

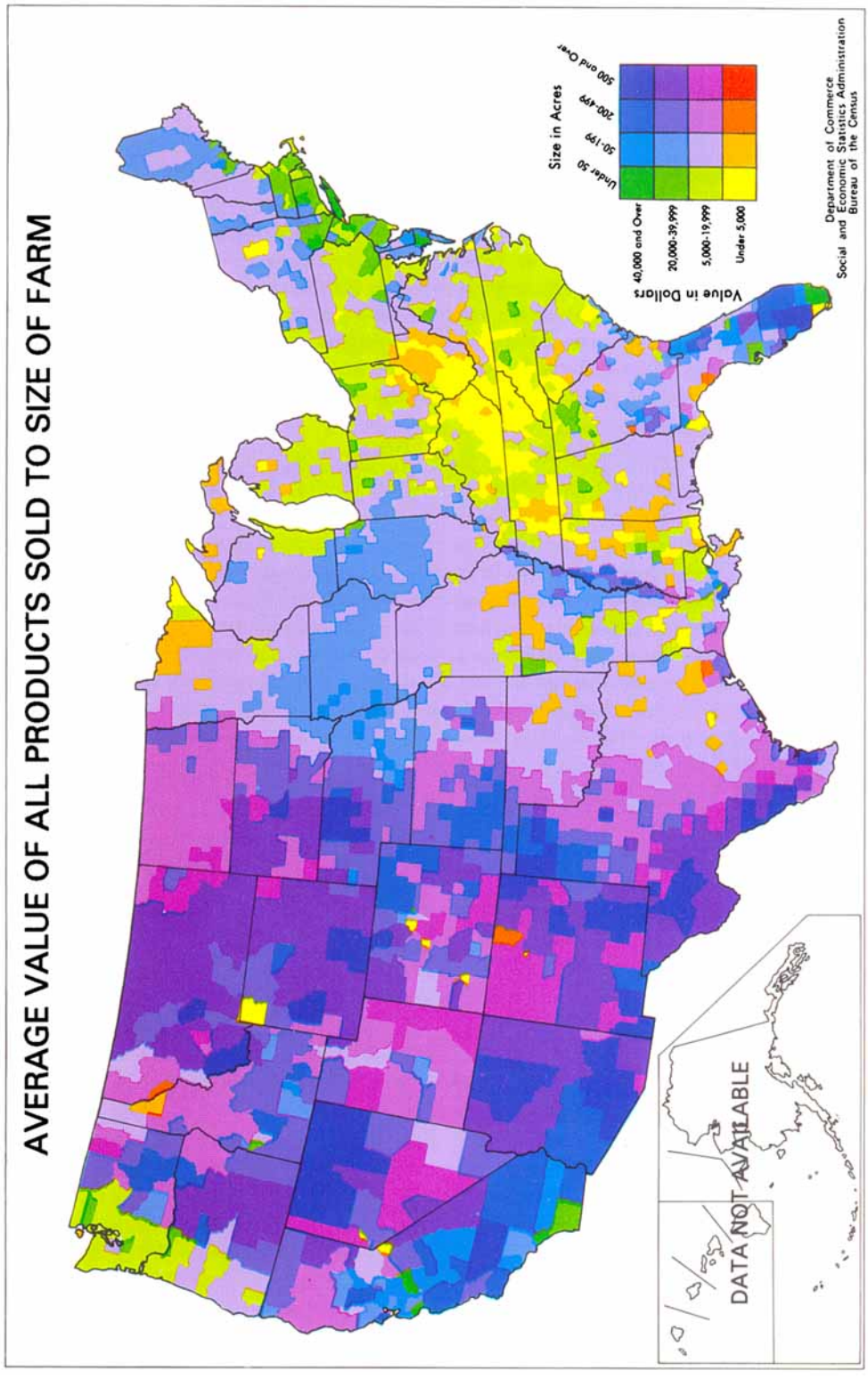

AVERAGE VALUE OF ALL PRODUCTS SOLD TO SIZE OF FARM
Size in Acres
Under 50
50-199
200-499
500 and Over
Value in Dollars
40,000 and Over
20,000-39,999
5,000-19,999
Under 5,000
Department of Commerce
Social and Economic Statistics Administration
Bureau of the Census
DATA NOT AVAILABLE

FIGURE 4

their own minds, randomly placing the colors. Many were basing at least some choices on connotative associations with the colors (dull colors with the dull outlook of little income or education, green with money, purple with academia, and so on) and many were at least attempting a coherent scheme in which neighboring tones related to one another in some way. Certain colors tended to give more difficulty than others, but the particular scheme that a subject was attempting to use seemed to have some effect on which colors were most difficult to place. Most important, perhaps, was that when shown the original scheme, most subjects *recognized* it as an organized one even if they had not thought of this spectral system or had not been able to achieve its arrangement. The exclamations of recognition on the part of several subjects were convincing that it was not simply a matter of trying to please the examiner, and not all who agreed that there was visual coherence in that arrangement agreed that it was better than their own (62% said yes, it was preferable).

In summary, Experiment I showed that map users do not spontaneously organize tones into the color scheme used on spectrally encoded two-variable maps, but they do recognize order in that arrangement. The legend on these maps, then, is necessary to convey *relative* values represented by the symbols as well as to convey the value levels represented, the latter being the normal function of the legend on a quantitative map.

Experiment II—Spatial Patterns

The second experiment involved a very different task than the first and required examination of symbols in spatial arrangement. But how does one evaluate the intelligibility of symbols in their spatial context? There is no obvious question that can be asked of subjects to reveal whether map pattern is intelligible, but it is quite possible to ask questions that do demand the mental processing of entire spatial patterns. Visually judging spatial complexity or correlation, for example, depends on the examination of symbols that are in spatial arrangement.

In the case of the present experiment it was visual correlation that was to be performed. The purpose was not to determine how well people can judge relationships from maps nor to discover the criteria used by subjects in such judging.[8] Rather, the experiment was intended to reveal whether subjects could process whole patterns of symbols on spectrally encoded two-variable maps. The concept of correlation was defined as clearly as possible to the participants in nontechnical terms so that variation in understanding of the concept would be at a minimum. Further, in order to have some basis for judging the results, the single-variable counterparts of all maps were tested as well as the two-variable versions.[9]

The maps used for this test were not actual distributions but were generated normal distributions placed within a constant 10 × 10 grid (Figs. 8 and 9). A series of patterns was chosen according to predetermined levels of correlation and autocorrelation, the latter a measure of relationship between neighboring values on the map that reflects something distinctly visible about the pattern.[10] The inclusion of combinations of correlation levels meant that various differences in correlation were also represented in the experiment. Both three-class and four-class maps were used, each with two types of class intervals—quantiles, in which there are equal numbers of observations in each category, and breaks at standard deviation units, chosen such that there was a large portion of the values in the intermediate class(es) and small portions in the extreme categories.

The three-class set included fifty-two triplets of distributions representing four levels of autocorrelation, several levels of correlation, several differences in correlation, and the two class-

[8] See, for example, Harold H. McCarty and Neil E. Salisbury, *Visual Comparison of Isopleth Maps as a Means of Determining Correlations Between Spatially Distributed Phenomena*, Department of Geography Studies, No. 3 (State University of Iowa: Iowa City, 1961); Phillip C. Muehrcke, "Visual Pattern Analysis—A Look at Maps," unpublished doctoral dissertation, University of Michigan, 1969; Judy M. Olson, "The Effects of Class Interval Systems on the Visual Correlation of Choropleth Maps," unpublished doctoral dissertation, University of Wisconsin, 1970; Jean-Claude Muller, "Associations in Choropleth Map Comparisons," *Annals*, Association of American Geographers, Vol. 65 (1975), pp. 403–13; R. E. Lloyd and T. R. Steinke, "The Decisionmaking Process for Judging the Similarity of Choropleth Maps," *The American Cartographer*, Vol. 3 (1975), pp. 177–84.

[9] Copies of sample test sets and complete instructions given to subjects are available from the author.

[10] Judy M. Olson, "Autocorrelation and Visual Map Complexity," *Annals*, Association of American Geographers, Vol. 65 (1975), pp. 189–204.

TABLE 1.—CHARACTERISTICS OF MAPS USED IN EXPERIMENT II

Variable	Number of levels	Description of levels
Correlation level (tau, using class values)	8	.0, .1, .3, .6, .7, .8, .9
Difference in correlation	5 (3-class maps) 1 (4-class maps)	.1, .2, .3, .6, .9, .3
Autocorrelation (tau, using class values)	4	.0, .3, .6, .9
Class interval method	2	quantiles, standard deviations ($\pm$1s, or 0 and $\pm$1s)
Number of classes	2	3, 4
Type of map	2	Separate maps (black and white), spectrally encoded two-variable maps

interval methods (Table 1). Only a subset of the fifty-two triplets was used in the case of the four-class maps because of printing considerations; all four-class maps had to be screened simultaneously in order that colors would be consistent and matchable with a common legend. The four-class maps used were those falling in the critical area of correlation differences where correlation is neither very easy nor impossible to detect visually.

While the concept of correlation was the same whether subjects were working with single-variable or two-variable maps, the specific processing of the two map types was different. Subjects had to chose which of two maps was more correlated with a third (referent) map in the case of the single-variable maps (Fig. 8). In the case of the two-variable maps, the referent distribution was overprinted on the two choice distributions and the task was to choose which of the two maps, each showing two distributions, showed the two that were more highly correlated (Fig. 9).

Each subject performed the task with a booklet of twenty-four to twenty-six two-variable, color-coded map pairs and with a booklet of the black-and-white separate-map versions of the same distributions (presented in a different order). Results for the two methods could thus be directly compared with no doubt that the same population of subjects was drawn upon for the two sets of responses. Some subjects worked with the two-variable maps first and the single-variable maps second, whereas others worked with them in the opposite order, thus ensuring that order of presentation would not bias results. Between fifty-four and sixty-six subjects responded to each map set.

Directions were read aloud and page numbers were called at fifteen-second intervals, primarily for purposes of assuring that complete instructions were read, that all subjects were taking the test under approximately identical conditions, and that all maps were being examined for approximately equal lengths of time with no rush to finish on the part of those who might start at a slower pace under less controlled conditions. The regimented procedure also meant that the testing time (twenty minutes) was closely predictable, an important consideration when class time or volunteer time is involved in an experiment.

While the test design included several variables, the results of primary interest here are those concerning the effects of the mapping method. By using identical variations on the many other variables, these other variables were controlled for purposes of examining the differences with respect to mapping method. The average difference between the proportion of subjects answering incorrectly on the single-variable maps and the proportion answering incorrectly on the two-variable maps was 9% and the Z-value of 6.10 is significant at the .001 probability level (n = 126 paired sets of maps). In other words, it is very unlikely that a difference this large would have arisen by chance if there were no difference between the two methods with regard to this task.

While these results indicate that under the conditions of the experiment the performance with the two-variable maps was indeed poorer, a close examination of detailed results provides some insight that prevents the naive conclusion that these maps are simply not as effective for judging correlation (and presumably for processing symbols in their spatial context) as the conventional separate displays of the distributions. When results for individual subjects are examined it can be determined whether each in-

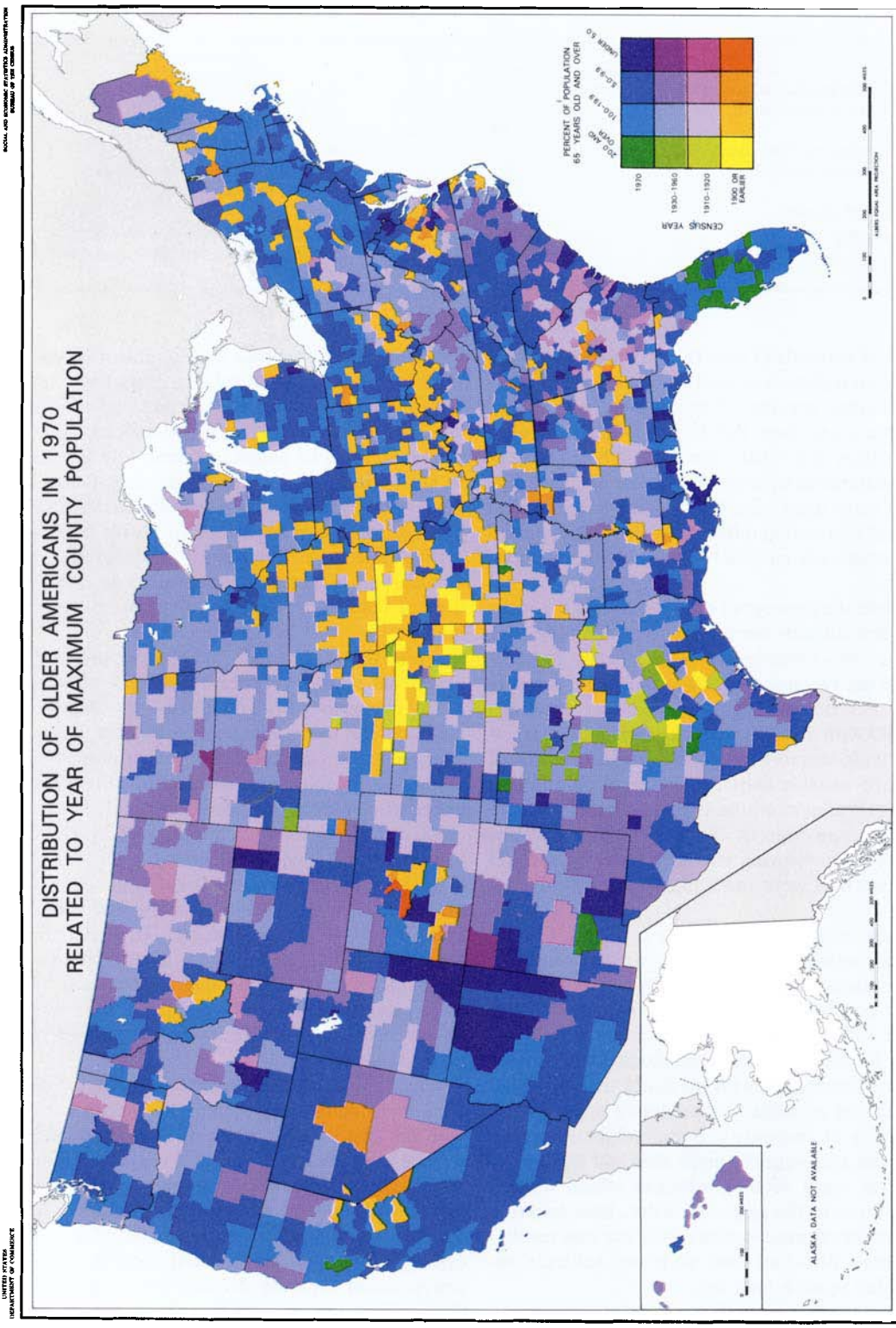

DISTRIBUTION OF OLDER AMERICANS IN 1970
RELATED TO YEAR OF MAXIMUM COUNTY POPULATION
PERCENT OF POPULATION 65 YEARS OLD AND OVER
20.0 AND OVER
15.0-19.9
10.0-14.9
LOWER 5.0
CENSUS YEAR
1970
1930-1960
1910-1920
1900 OR EARLIER
ALASKA: DATA NOT AVAILABLE

FIGURE 5

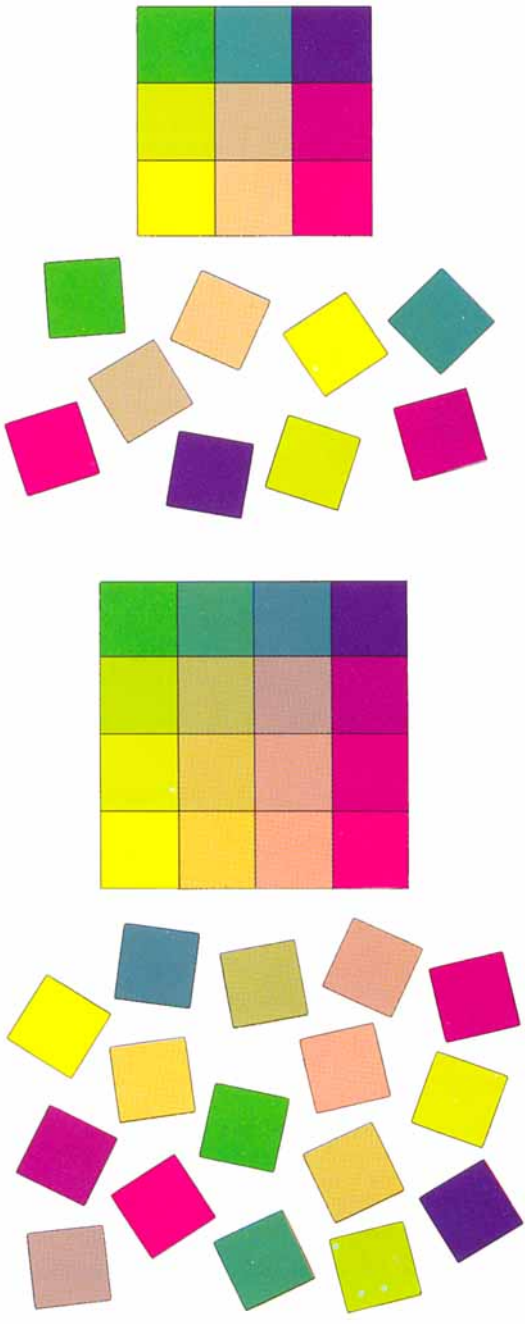


FIG. 6. The sets of colors used in the arrangement task of Experiment I. Each subject worked with only one of the color sets. Screen percentages used in the three-by-three set were 100, 30, 0 cyan for the top through bottom rows; 100, 50, 0 yellow and 0, 30, and 100 magenta for the columns from left to right. Percentages for the four-by-four set were 100, 50, 20, and 0 cyan for rows; 100, 70, 40, 0 yellow and 0, 20, 50, and 100 magenta for columns. Below each block is a scatter of the colors as they might have appeared for a subject faced with the arrangement task.

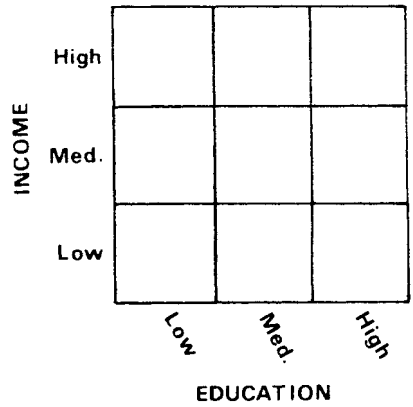


FIG. 7. The grid into which subjects working with nine colors were to place their color chips.

dividual was 1) guessing (answering approximately one-half correctly, since there are only two choices in each case); 2) answering correctly; or 3) giving consistently incorrect answers according to the criteria given them.

If one assumes guessing and places a 95% confidence interval about the expected value of .5, then any scores outside of that interval can reasonably be considered as nonguessing scores. With the black-and-white maps, 86% of the 295 participants were answering correctly at better than guessing level, only 14% were in the interval within which the guessing hypothesis would not be rejected, and none were giving consistently incorrect answers. In the case of the color maps, 58% were answering correctly at better than guessing level, 33% were within the guessing range, and 9% were giving consistently incorrect answers. Interestingly enough, if we then compare the scores on the single-variable versus the two-variable booklets for *those who were doing better than guessing on the color set* (in other words, for those who had actually "caught on" to the technique for judging), we find that 58% of them had better scores on the color maps. This is not an overwhelming figure (a hypothesis of 50% could not be rejected at the .05 level) but it raises the question of whether the two types of map fall in opposite order once the subject passes over the cognitive hurdle of seeing how the task is to be accomplished. At the very least it is evidence that the two-vari-

TABLE 2.—SUMMARY OF GROUPS: EXPERIMENTS III AND IV

Group	Map	Explanatory note	Number of subjects	Course and school of subjects[1]
1	Average Value of All Products Sold to Size of Farm	Yes	19	Introductory cartography, Boston University
2	Same as group 1	No	14	Freshman psychology, Boston University
3	Distribution of Older Americans in 1970 Related to Year of Maximum County Population	Yes	26	Freshman psychology, Boston University
4	Same as group 3	No	20	Oceanography class, Boston University
5	Interrelationship of Educational Attainment and Per Capita Income	Yes	20	Introductory cartography, Boston University
6	Same as group 5	No	19	Freshmen psychology, Boston University
7	High School Graduates as a Percent of Persons 25 Years Old and Older, 1970; Per Capita Income, 1970 (two separate maps)	No	25	Physical geography, Boston State College (8); freshman psychology, Boston University (7); introductory geography, Salem State College (10)

[1] All subjects were students. None were experienced in using two-variable maps, nor were any color-blind to their knowledge. No student participated more than once.

able maps are not associated with poorer performance by this group of subjects.

In summary, Experiment II used the task of visual judgment of correlation to assess the ability of users to process symbols in spatial context. Under the conditions of the experiment, subjects performed considerably better with the separate maps and many were guessing or consistently choosing the wrong answer with the two-variable versions. Those who caught on to the appropriate judging technique for the color maps, however, did perform slightly better with them than with the separate maps, although the difference was not statistically significant.

Experiments III and IV—Real Maps: Information Extraction and Attitudes

The third and fourth tests were performed by the same subjects and utilized the same maps; therefore, they are grouped together here for clarity of exposition even though they were very different from one another in nature and purpose. The third experiment, "Part 1" of the activities performed by this set of subjects, was to explore what sorts of information are extracted from spectrally encoded two-variable maps, that is, from full maps with title and legend and realistic pattern. The fourth experiment, "Part 2" of the activities of this set of subjects, explored what these people thought of the maps, or their attitudes toward them.[11]

Subjects in this pair of experiments were tested in groups of seven to twenty-six. Each was given a two-variable map or the pair of separate maps of education and income and written instructions of the following form for Experiment III:

> Please examine the attached map and write in the space below *what this map tells you,* or *what you learn from this map* about the distribution of value of products sold and size of farm. There will be a time limit of 10 minutes.

The written answers were set aside after the ten

[11] For other cartographic studies dealing with attitudes, see: Barbara Bartz Petchenik, "A Verbal Approach to Characterizing the Look of Maps," *The American Cartographer,* Vol. 1 (1974), pp. 63–71; Borden D. Dent, "Communication Aspects of Value-by-Area Cartograms," *The American Cartographer,* Vol. 2 (1975), pp. 154–68; and Patricia P. Gilmartin, "Evaluation of Thematic Maps Using the Semantic Differential Test," *The American Cartographer,* Vol. 5 (1978), pp. 133–29. The word pairs used in Experiment IV are particularly similar to those of Dent.

TABLE 3.—MEDIANS AND INTERQUARTILE RANGES ON WORD PAIRS[1]

	Products/farm size		Older pop./year max		Education/income		Separate educ/inc maps
	w	*wo*	*w*	*wo*	*w*	*wo*	(*wo*)
Visually displeasing/visually appealing	8 (2)	8 (3)	7½ (2)	7 (4)	9 (3)	8 (3)	8 (1)
Not informative/informative	7 (3)	8 (3)	7 (4)	6½ (1)	8 (5)	7 (3)	8 (2½)
Worthless/valuable	7 (3)	6½ (4)	6 (2)	6 (3)	7 (4)	6 (3)	7 (3)
Misleading/truthful	5 (3)	6 (3)	7 (4)	5 (4)	5 (4)	5 (3)	5 (3)
Bad/good	6 (2)	5 (3)	6½ (3)	5 (2)	7 (4)	5 (1)	6 (2½)
Lacking authority/authoritative	7 (2)	5 (1)	5½ (3)	5 (1½)	8 (4)	5 (1)	5 (3½)
Generalized/specific	7 (3)	6 (3½)	7 (4)	6½ (6½)	5 (4)	4 (4)	5 (4)
Difficult to read/easy to read	6 (6)	5½ (4)	3½ (2)	3 (3½)	7 (6)	4 (3)	9 (2)
Conventional/innovative	8 (2)	5 (2½)	6 (3)	7 (4)	8 (2)	5 (2)	5 (3½)
Does not communicate/communicates well	7 (3)	7 (3)	5 (2)	5½ (3½)	8 (4)	6 (4)	8 (3)
Does not form an immediate impression/ does form an immediate impression	7 (9)	6 (3)	5 (5)	5 (4)	8 (4)	6 (5)	8 (5)
Usual/unusual	8 (3)	5½ (4)	7 (3)	7 (3½)	7 (4)	5 (4)	5 (2½)
Does not show individual distributions clearly/shows individual distributions clearly	7 (3)	5½ (6)	6 (5)	4½ (5)	6 (5)	5 (5)	6 (5½)
I cannot judge the closeness of relationship . . . /I can judge . . .	5½ (5)	6 (4)	5 (3)	5 (3)	7 (2)	6 (4)	9 (4)
The distributions are not interesting/ . . . are interesting	8 (3)	5½ (5½)	6 (2)	6 (3)	9 (3)	6 (5)	8 (3½)
The pattern shown is spatially complex/ . . . is spatially simple	4½ (4)	5½ (4)	4 (3)	3 (3½)	5 (5)	5 (3½)	6 (3)
The method of mapping the data is not interesting/ . . . is interesting	8 (2)	5 (3)	6½ (2)	6½ (3)	9 (2)	5 (2)	7 (3)
The method of mapping the data is complex/ . . . is simple	5 (5)	4 (3)	4 (3)	4 (3)	5 (4)	4 (3)	6 (4)
This map does not make me want to learn more about maps/ . . . makes me want . . .	7 (3)	4 (3)	5 (4)	5 (2)	6 (4)	5 (5)	5 (3)
It is difficult to match the colors . . . / it is easy . . .	8 (3½)	5 (4)	5 (6)	2½ (4)	7 (7)	5 (9)	10 (3)
The relationship . . . is negative . . . / . . . is positive . . .	8 (3)	5 (3)	5 (3)	5 (3)	7½ (5)	6 (5)	8 (3)

[1] Interquartile range is shown in parentheses; *w* indicates *with* explanatory note, *wo* is *without* explanatory note.

minutes and the forms were distributed for Experiment IV. These forms contained a series of polar word pairs, such as "visually displeasing" and "visually appealing" (Table 3). For each of these word pairs, subjects were to rate the maps on a scale of zero to ten.[12]

[12] The exact wording of the instructions was as fol-

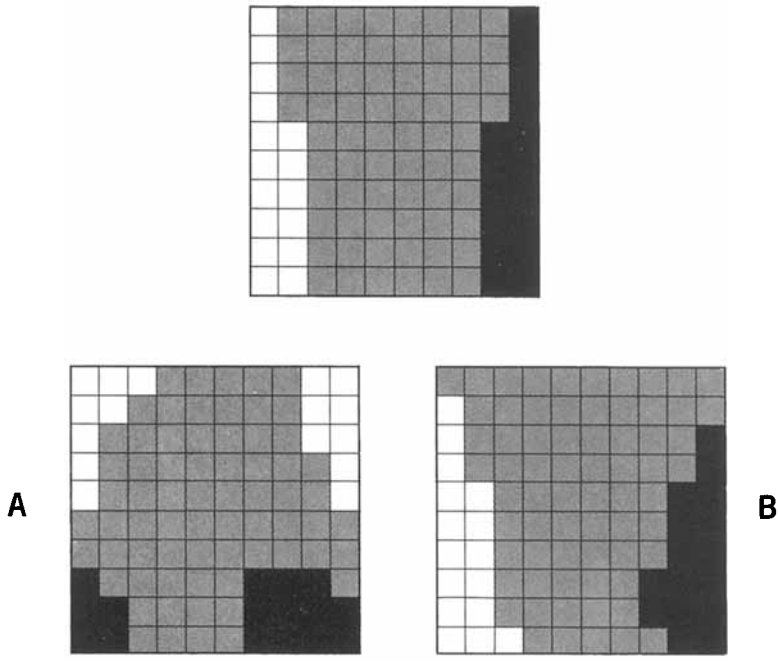


FIG. 8. An example of a black-and-white test item (separate maps of each distribution). Subjects had to choose which map, A or B, was more similar to the upper (referent) map. "Similarity" was clarified in the instructions to mean locational correspondence of values.

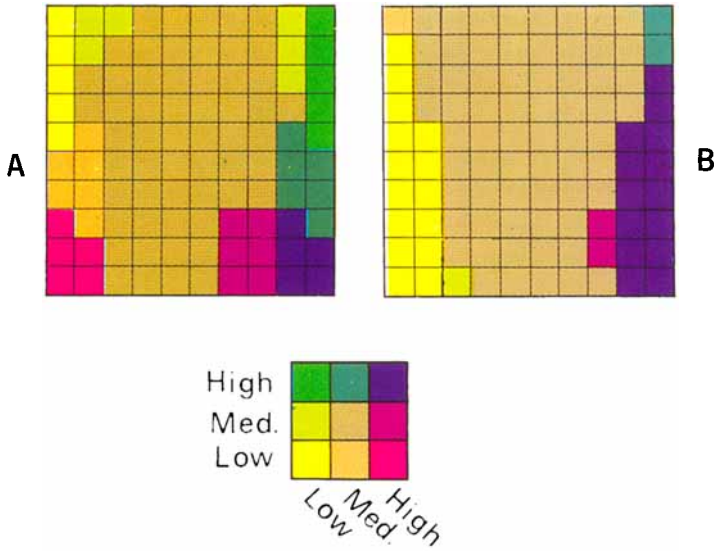


FIG. 9. An example of a test item from a color booklet. This item is the two-variable counterpart of the maps in Fig. 8; the referent map has been overlaid on each of the choice maps. Subjects had to choose which map, A or B, showed the two distributions that were more similar to one another. The legend was always visible with these test items.

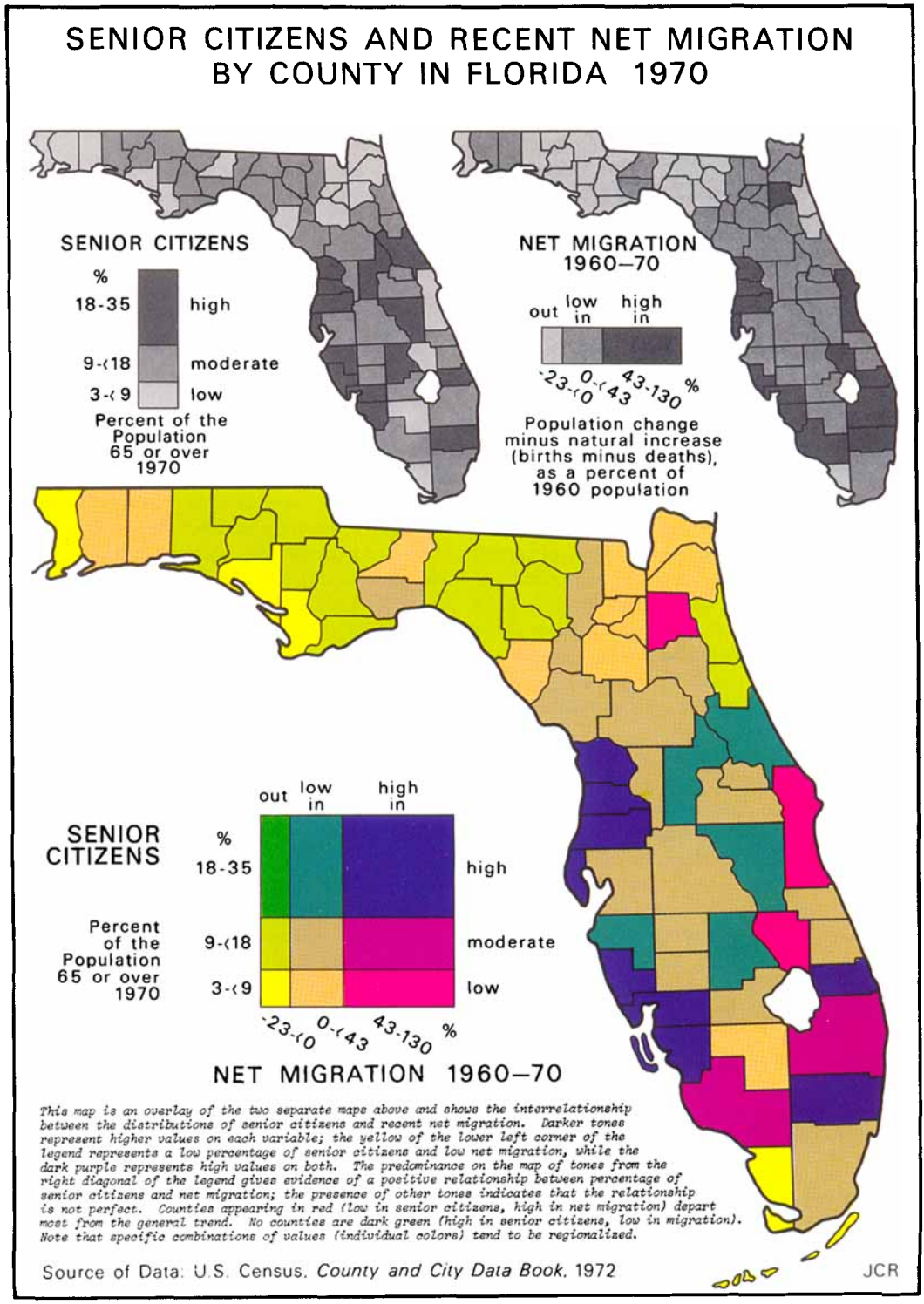


FIG. 10. A map display incorporating a prominent legend, the separate maps as well as the composite two-variable version, and explanatory material. (Adapted from a map by Jonathan Corson-Rikert.)

The subjects working with each composite two-variable map in Experiments III and IV were further subdivided into two groups; one was given only the map itself, the other was given a short explanatory note as well as the map (Table 2). Thus there were six groups working with the composite maps. A seventh group worked with the separate maps of the education and income distributions (Figs. 2 and 3, no explanatory note). Inclusion of this seventh group allowed results for separate maps to be compared directly with those for the compositely printed two-variable version of the same distributions.

The results of Experiment III, with its subjective and open-ended question, cannot be treated with statistical rigor. But degree of objectiveness in test questions is not a very reliable measure of the importance of the information obtained from them. In this case the results from the open-ended question offer some desired information that simply cannot be obtained through tasks that are subject to rigorous statistical analysis. The underlying issues were whether people obtain information from these maps without their attention being directed to specific parts of the map or to specific relationships portrayed, and if they do extract information, what is its nature?

The participants' written answers were read for a general sense of what was expressed and counts were then made of the frequency of mention of such items as regions (the South, Middle West, etc.), specific states, color names, relationships, trends, average values, individual distributions, and difficulty of reading. Statements of obvious error (misinterpretation) were also noted.

The most striking observation from this experiment is that these subjects were in no way overwhelmed by the two-variable maps. While critics have often dismissed them as too complex and unable to convey the separate distributions, these subjects had no preconceived notions of what they should be seeing and approached them at face value. No blank answer sheets were submitted and relatively few complaints about the difficulty of the maps were included in the essays.

Generally, references to regions and states abounded. While colors were referenced by a number of subjects, more often it was the value combination (such as low education, low income) represented by the color that was mentioned in connection with these regions, reflecting a sort of transparency of the symbol system to the values represented. In other words, people were not simply describing the form of the map, but were using that form to access the content of the map. Relationships were less often mentioned but were more common with the groups who were given the explanatory note, that note drawing attention explicitly to the concept of relationship. In the case of the education/income map, relationships were mentioned about as frequently without the explanation as with it, presumably because people already strongly associated the two variables. References to spatial trends were extremely difficult to extract from the comments except in the case of the Value of Products and Farm Size map when several such references appeared. Average values and individual distributions were infrequently referenced. Clear and direct complaints of difficulty were common only in the case of the Older Americans and Year of Maximum Population map without the explanatory note. Serious errors of interpretation of what was portrayed occurred in a few cases with the Farm map but were common only with the Older Americans and Year of Maximum Population map. With the latter map eighteen out of twenty subjects who were not given the explanatory note clearly revealed misunderstanding; fifteen out of twenty-six of those provided the note (and whose attention was drawn orally to the distributions portrayed) were also clearly confused about them.

The seventh group, which worked with the individual Education and Income maps, commonly referenced regions and states, and relationships were more frequently mentioned. Trends and average values still received little attention. Combinations of values were mentioned with these maps, but references to homogeneous combinations were far fewer than elicited by the two-variable map. The essays on the single-variable maps were not no-

lows: "Please consider each word pair below and rate the map on the 0 to 10 scale. For example, if you feel the map is visually appealing, place a mark at the high end of the scale; if it is visually displeasing, place the mark at the low end; and if you feel it is neither appealing nor displeasing, place the mark at 5. (If you feel it is slightly appealing or displeasing, place the mark to the right or left of five accordingly.)"

ticeably better in quality (accuracy of information) nor were they longer in length (amount of information).

In summary, Experiment III revealed that these potential users can and do obtain information from spectrally encoded two-variable maps, primarily about regions of homogeneous combinations of values. Content and form do interact with one another as shown by the inaccuracies among the comments of those working with the Older Americans and Year of Maximum Population map.

The basic results of Experiment IV are the average (median) range of subjects on the various attitude scales (Table 3). Among the most interesting observations is that, in general, subjects reacted positively to the aesthetic quality of the two-variable maps (visually displeasing/visually appealing), but the Older Americans and Year of Maximum Population map was not rated quite as highly as the other two maps (average rating approximately 7 as opposed to 8 or 9).

Another striking observation is that median ratings on many word pairs decreased by at least two units on the 0–10 scale when no explanatory note was provided to subjects. The Products Sold and Farm Size map without the explanatory note was rated less authoritative, less innovative, and less unusual than the same map *with* explanatory note. Also, both the distribution and mapping method were rated less interesting and subjects were less inspired to learn more about maps. The group without the explanatory note found the distributions less related to one another and they even found it more difficult to match the colors with those in the legend!

Similar decreases in ratings were apparent when the Education and Income map was presented without the explanatory note, although there were differences in the word pairs on which these differences occurred. The effect of the explanatory note was least with the Older Americans and Year of Maximum Population map, for which only two median ratings decreased by two or more units when the explanatory note was not available. With this map the ratings when the explanatory note was included were already somewhat lower than for the other two maps (median of 6, compared to 7 on the other two).

The separate maps of education and income elicited several median ratings that were higher than those for the composite map of these distributions without the explanatory note, but when compared to the composite with the note, some changes were positive and others negative. The separate maps were thought to be less authoritative, less innovative, less unusual, and mapped with a less interesting method; but they were judged easier to read, closeness of relationship was considered easier to estimate, and colors were rated as easier to match.

In sum, Experiment IV showed positive reactions to the aesthetics of spectrally encoded two-variable maps. Attitude ratings were more positive when the short explanatory note was included and when the logical connection between the variables was understood. Readability was rated higher for the separate maps than for the corresponding two-variable map but interest variables were rated lower.

Conclusions

Given the results of these four experiments what can we conclude about spectrally encoded two-variable maps? The experiments bear evidence that it is information about regions of homogeneous value combinations that is readily available, and generally the user is not unduly confused by such a map nor adversely affected in attitude. Perhaps more important, however, are the implications of the study for the design of the two-variable map display.

The revised design of the display would include several features (Fig. 10):

1) A very prominent and clear legend is necessary. This was suggested by Experiment I, the legend arrangement exercise, which illustrated the critical role of the legend in conveying even relative values.

2) Both the single-variable and two-variable versions of the distributions should be shown. This was suggested by Experiment III, the essays on the real maps, in which information extracted from the two-variable maps differed in *nature*, rather than in quantity or accuracy, from information extracted from single-variable maps. The display of both versions, then, may be the key to increasing or improving extraction of information by the user. The separate maps should probably be displayed in black and white, rather than in the component colors of the composite map, so that extraction of spatial information will be encouraged rather than informa-

tion about how colors combine to produce other colors.

3) Explanatory notes should be included to reinforce the confidence of the user and to draw attention to the types of information extractable from the display. This feature is suggested by Experiment IV, in which more positive responses were elicited when such explanatory material was present.